This is your third homework assignment. I strongly encourage you to typeset it using the LaTeX template provided. It is due by ~~Friday, January 31st~~ Monday February 3rd.

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(1) *(matrix multiplication).* This question will introduce you to three different ways of thinking about matrix multiplication. For all three parts, let $U \in \mathbb{R}^{m\times n}$, $V \in \mathbb{R}^{m\times r}$ and $W \in \mathbb{R}^{r\times n}$. By $\mathbf{u}_i$ (respectively $\mathbf{v}_i$ and $\mathbf{w}_i$) I mean the i-th column of U (respectively V and W).

(a) Show that $U^\top V = \begin{bmatrix} \mathbf{u}_1^\top\mathbf{v}_1 & \mathbf{u}_1^\top\mathbf{v}_2 & \dots & \mathbf{u}_1^\top\mathbf{v}_n \\ \vdots & \ddots & \ddots & \vdots \\ \mathbf{u}_n^\top\mathbf{v}_1 & \mathbf{u}_n^\top\mathbf{v}_2 & \dots & \mathbf{u}_n^\top\mathbf{v}_n \end{bmatrix}$ *(Recall that $\mathbf{u}_i^\top\mathbf{v}_j$ is the dot product between $\mathbf{u}_i$ and $\mathbf{v}_j$)*

(b) Show that $UW^\top = \sum_{i=1}^{n} \mathbf{u}_i\mathbf{w}_i^\top$ *(Recall that $\mathbf{u}_i\mathbf{w}_i^\top$ is the outer product of two vectors, and will give a rank one $m \times r$ matrix.*

(c) Show that $U^\top V = [U^\top\mathbf{v}_1, U^\top\mathbf{v}_2, \dots, U^\top\mathbf{v}_r]$.

General Hint: For all three parts above, you have to show equality of two matrices. So, you probably want to work out the (i,j)-th entry of the left hand side, and the (i,j)-th entry of the right hand side, and show that they are equal.

(2) *(k-means)* In this question you will work through a single iteration of k-means for a simple data set. Let $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_{10}\} \subset \mathbb{R}$ where:

$$\mathbf{x}_1 = -0.82, \quad \mathbf{x}_2 = -1.33 \quad \mathbf{x}_3 = -3.63, \quad \mathbf{x}_4 = -1.62, \quad \mathbf{x}_5 = -2.95$$
$$\mathbf{x}_6 = 0.95, \quad \mathbf{x}_7 = 0.53, \quad \mathbf{x}_8 = 1.79, \quad \mathbf{x}_9 = 0.95 \text{ and } \mathbf{x}_{10} = 2.28$$

We will take $k = 2$, that is we are seeking to partition $\mathcal{X} = \mathcal{X}_1 \cup \mathcal{X}_2$. Suppose that the partition vector, Π, is randomly initialized to $\Pi^{(0)} = [2,1,1,1,2,2,1,2,1,2]^\top$ *I'm transposing to keep up with the convention that our vectors are column vectors. The* 0 *superscript indicates that this is the initial partition vector.*

(a) Using $\Pi^{(0)}$, determine $\mathcal{X}_1^{(0)}$ and $\mathcal{X}_2^{(0)}$.
(b) Now, compute the centroids $\mu_1^{(0)}$ and $\mu_2^{(0)}$.
(c) Now, compute $\mathcal{X}_1^{(1)}$ and $\mathcal{X}_2^{(1)}$.
(d) Finally, compute $\Pi^{(1)}$.

Optional: Check your answer using Python.

(3) *Non-negative matrix factorization* Suppose that you run a small grocery store, and you wish to understand your customers better. Specifically, you'd like to *segment* your customers into a small number of groups with similar purchasing characteristics. So, you do the following:

(a) You identify 6 main categories of product: Bread, Fresh fruit & Veggies, Cigarettes, Milk, Canned Food, Newspapers.
(b) You select 5 customers and (legally and ethically!) track their purchasing habits for a month. Specifically you record the amount that they spend on each category of product in the month, and then normalize this amount to a number between 0 and 1.

This data is presented in matrix X in (**??**). The first column corresponds to customer 1, the second to customer 2 and so on. The first row corresponds to Bread, the second to Fresh fruit & Veggies and so on. Also presented in (**??**) is the *non-negative matrix factorization* of X. That is, $X \approx WH$.

(1)

$$X = \begin{bmatrix} 0.735 & 0.179 & 0.079 & 0.698 & 0.307 \\ 0.093 & 0.655 & 0.655 & 0.436 & 0.604 \\ 0.686 & 0.782 & 0.547 & 0.427 & 0.471 \\ 0.138 & 0.223 & 0.573 & 0.846 & 0.164 \\ 0.427 & 0.018 & 0.488 & 0.193 & 0.819 \\ 0.78 & 0.69 & 0.354 & 0.221 & 0.184 \end{bmatrix} \quad W = \begin{bmatrix} 0.162 & 0.449 & 0.696 \\ 0.784 & 0.0 & 0.0 \\ 0.632 & 0.599 & 0.0 \\ 0.413 & 0.0 & 0.648 \\ 0.62 & 0.0 & 0.0 \\ 0.336 & 0.766 & 0.0 \end{bmatrix} \quad H = \begin{bmatrix} 0.268 & 0.557 & 0.896 & 0.538 & 0.854 \\ 0.94 & 0.595 & 0.0 & 0.0 & 0.0 \\ 0.226 & 0.0 & 0.0 & 0.917 & 0.0 \end{bmatrix}$$

(a) Recall that the columns of W represent the categories, or segments, of customers. Interpret these segments based on the products they purchase the most.

(b) With which segment is Customer 1 most strongly associated?

(c) With which segment is Customer 3 most strongly associated?

(4) *Spectral Clustering*

For this question let A denote the adjacency matrix of a graph G. Let $L = D - A$ denote the Laplacian and let $\mathbf{I}_s = \begin{cases} \sqrt{\frac{|S^c|}{|S|}} & \text{if } v_i \in S \\ -\sqrt{\frac{|S|}{|S^c|}} & \text{if } v_i \notin S \end{cases}$ for any $S \subset V$.

(a) Show that $L\mathbf{1} = 0$ where $\mathbf{1}$ denotes the all-ones vector.

(b) Suppose that G has two connected components: $V = S_1 \cup S_2$. Show that $L\mathbf{1}_{S_1} = 0$ where: $(\mathbf{1}_{S_1})_i = \begin{cases} 1 & \text{if } i \in S_1 \\ 0 & \text{otherwise} \end{cases}$. *(Of course, one can similarly show that $L\mathbf{1}_{S_2} = 0$)*

(c) It is a fact that $\mathbf{v}^\top L\mathbf{v} = \frac{1}{2}\sum_{i,j} A_{ij}(v_i - v_j)^2$ for any $\mathbf{v} \in \mathbb{R}^n$ *(You do not need to prove this!)*. Use this to show that:

$$\text{Rcut}(S) = \frac{1}{n^2}\mathbf{I}_S^\top L\mathbf{I}_S$$